



Full length article

Life cycle assessment shows admixtures support high-performance low-carbon LC³ concretePippa Edwards^a, Sébastien Dhers^b, Rupert J. Myers^{a,*}^a Department of Civil and Environmental Engineering, Imperial College London, SW7 2AZ, London, UK^b Master Builders Solutions Global Research Center/Master Builders Solutions Deutschland GmbH, Dr-Albert-Frank-Strasse 32, 83308 Trostberg, Germany

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ABSTRACT

Limestone calcined clay cements (LC³) have high potential to reduce concrete's carbon footprint. However, the impact of admixtures on carbon footprint of LC³ concrete via performance enhancement is not well understood. This paper investigates the carbon footprint, compressive strength performance, and carbonation resistance of CEM II/A-LL, LC³-50, and LC³-35 concretes dosed with admixtures (polycarboxylate ether-based superplasticizer, C-S-H nanoparticle accelerator, silane-based water repellent). Generally, the results show that accelerator can increase the strength and carbonation exposure classes of LC³ concretes. We show that LC³-50 and LC³-35 concretes reduced CO₂-eq. emissions by 22 – 38 % compared to CEM II/A-LL, and that admixtures can contribute 13% of their cradle-to-gate CO₂-eq. emissions. Optimal admixture addition and clinker content depends on the strength and carbonation exposure classes required, but of the concrete types studied, the optimum combinations always contained LC³ and not CEM II/A-LL. Admixtures support the production of high-performance concretes with low carbon footprint.

1. Introduction

Cement production is responsible for ~ 8% of global anthropogenic CO₂ emissions. Without action, the carbon footprint of the concrete industry is projected to emit 3.8 Gt of CO₂ per year by 2050 (GCCA, 2021). Emissions from cement production are dominated by Portland cement clinker (hereafter clinker) production, and reducing the clinker content in cement is a proven strategy for cement decarbonisation (GCCA, 2021; Miller et al., 2016; Scrivener et al., 2018b; Shah et al., 2022). Limestone calcined clay cement (LC³) has received significant attention (e.g., (Scrivener et al., 2018c, 2018a; Cancio Díaz et al., 2017; Scrivener et al., 2022; Hanein et al., 2021)) due to the wide availability of required raw materials, economic feasibility, lower carbon footprint, and good technical performance. Currently, the technical performance of LC³ cement allows further clinker reduction compared to other clinker substitutes (e.g., limestone, natural pozzolans (Edwards et al., 2026a)). For example, in mortar tests, LC³-50 (e.g., 50% clinker content) had comparable 7-day compressive strength to mortar containing 90% clinker cement (Scrivener et al., 2018c).

Despite this, concerns for the performance of very low clinker (e.g., <50% clinker content) LC³ concrete have been reported, including

workability and slump retention (e.g., (Zaribaf et al., 2015; Ferreira et al., 2017; Akhlaghi et al., 2017; Nair et al., 2020; Dhers et al., 2024)), particularly for pumped concretes (Olivo et al., 2025)), strength development (e.g., (Avet and Scrivener, 2018; Sun et al., 2024; Zunino and Scrivener, 2025)), and carbonation resistance (e.g., (Scrivener et al., 2018c; Bucher et al., 2015; Khan et al., 2019; Shah et al., 2015)). LC³ concretes are susceptible to high carbonation rates compared to high clinker concrete, as there is less CaO available to bind with CO₂ (Scrivener et al., 2018a). Carbonation reduces the durability of reinforced steel concretes, as carbonated concrete has lower pH which increases the risk of embedded steel corrosion when the carbonation front is sufficiently deep. For example, Khan et al. (2019) found that carbonation resistance decreases as clinker content decreases, and that LC³ concrete with 60% clinker content would require a 26% increase in concrete cover to achieve 50 year durability under outdoor exposure conditions to meet the same performance as ordinary Portland cement (OPC) concrete (Khan et al., 2019). Scrivener et al. (2018) reported that the carbonation rate of LC³ concrete is highly correlated with the relative humidity of the environment, as slow carbonation of concrete occurs at very low relative humidity (e.g., where there is insufficient moisture to form carbonic acid (Saetta et al., 1993)) and very high

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relative humidity (e.g., where water saturated pores block the diffusion of CO₂ (Papadakis et al., 1991a; Saetta et al., 1993)) (Scrivener et al., 2018a). Hence, solutions are needed to improve the performance of low-clinker LC³ concretes and support their application across a wide range of environmental conditions.

These performance concerns can be addressed using chemical admixtures (hereafter admixtures). Admixtures are a variety of products, including water-reducing and slump retaining agents (e.g., superplasticizers), accelerators, and water repellents, used to improve the properties of fresh and hardened concrete (Cheung et al., 2018; Plank et al., 2015). Superplasticizers, such as polycarboxylate ethers (PCE), disperse cement particles in wet concrete to improve the rheology of the mix. Previous work has investigated the impact of different superplasticizers on LC³ concretes (Dhers et al., 2024), and identified that specialised superplasticizers with delayed release mechanisms are required to overcome the higher specific surface area introduced by calcined clay. Accelerators are designed to reduce the setting time and increase early-age strength. Promising accelerators for LC³ include C-S-H nanoparticles, which provide a nucleation site for C-S-H crystal growth during early hydration (Thomas et al., 2009), and have been demonstrated to increase 1-day compressive strengths of LC³-50 by 45–100% (Morales-Cantero et al., 2023). Superplasticizers and accelerators also improve carbonation resistance, by allowing lower water to binder ratios for the same workability performance, and refining the pore structure of the concrete (Fu et al., 2023). As concrete carbonation requires capillary moisture, water repelling admixtures, such as those based on silanes, offer additional protection against carbonation by forming a hydrophobic film at the pore surface (Falchi et al., 2013; Zhu et al., 2013; Zhang et al., 2025). Therefore, the performance modifications offered by admixtures could enable high-performing, low clinker concretes.

However, admixtures have been largely overlooked in life-cycle assessment (LCA) of concrete. Marceau and Vangeem (2007) excluded admixtures from their life cycle inventory of Portland cement concrete (Marceau et al., 2007), claiming that admixtures are typically dosed <1 wt.% in the final product and do not contribute significantly to toxic emissions. Flower and Sanjayan (2007) reported the CO₂-eq. emissions associated with different admixture production, but only considered electricity consumption in their inventory (Flower and Sanjayan, 2007). They therefore concluded that admixture addition has negligible contributions to concrete emissions. Gursel et al. (2016) included the impact of superplasticizer in their study of concrete strength and durability performance with different supplementary cementitious materials (Gursel et al., 2016), using a ranking system to compare a durability indicator for chloride ingress with strength normalised carbon footprint of each concrete. More recently, inventory data for PCE superplasticizers have been reported based on experimental synthesis (Liu et al., 2021), and estimation based on stoichiometric calculations (Schiefer and Plank, 2023a). However, high-quality process-based inventory data supporting LCA of a wide range of admixture products are unavailable. Gursel et al. (2012) included data for several admixtures (e.g., superplasticizers, retarders, air-entrainers) in the GreenConcrete webtool, based on data from commercial Environmental Product Declarations (EPDs) (Gursel, 2012). Similarly, the OpenConcrete tool (Kim et al., 2022) and the Inventory of Carbon and Energy (ICE) database (Circular Ecology, 2024) also report EPD data for admixtures. Therefore, EPDs present the most comprehensive source of inventory data currently available for chemical admixtures. Moreover, the potential of using multiple admixtures to enable low clinker cement with acceptable technical performance has not been quantified within the LCA literature to date, since this analysis requires studies comparing similar mortar or concrete performances, such as workability and compressive strength.

The objective of this study is to investigate the environmental impact of chemical admixtures (superplasticizer, accelerators, and water repellents) on a reference CEM II/A-LL concrete and LC³ concretes at two clinker substitution rates (e.g., LC³-35, LC³-50). We combine the

strength and carbonation results with LCA to evaluate the carbon footprint of concretes on a performance basis, to understand the contribution of admixtures to concretes made with low-clinker composite cements, identify promising admixture dosing strategies, and demonstrate admixtures potential for producing low-carbon, high-performance concretes.

2. Methods

2.1. Materials

2.1.1. Cements

This study investigated three cements in seven concrete mixes: a reference CEM II/A-LL concrete using a CEM II/A-LL 42.5R Portland cement (85.8 wt.% clinker, 9 wt.% limestone, 5.2 wt.% gypsum), three LC³-50 concretes using LC³-50 cement (49 wt.% CEM I Portland cement, 32.4 wt.% calcined clay, 16.2 wt.% additional limestone, 1.9 wt.% additional gypsum) and three LC³-35 concretes using LC³-35 cement (33.9 wt.% CEM I Portland cement, 42 wt.% calcined clay, 21.1 wt.% additional limestone, and 2.9 wt.% additional gypsum). Note that ‘additional’ gypsum and limestone is in addition to gypsum and limestone contained in CEM I Portland cement per Table A1. Both LC³-50 and LC³-35 cements had a 2:1 calcined clay to limestone ratio and used a CEM I 52.5 R cement (90.1 wt.% clinker, 4.4 wt.% limestone, 5.6 wt.% gypsum) to simulate Portland clinker. The Portland cements were sourced from SCHWENK Zement GmbH & Co. KG, Germany, the calcined clay was sourced from Argeco, France, and limestone and gypsum were sourced from sh-minerals GmbH, Germany. The clay was processed via flash calcination, with a R³ test (Avet et al., 2016) value of 465 J/g as measured by calorimetry according to ASTM-C1897. Chemical analyses (Table A1) of all cementitious materials were determined by X-ray diffraction (XRD), using a D8 X-ray diffractometer from Bruker and an internal standard (Alpha-alumina powder IMS-135 from Independent Mineral Standards, Australia). Particle size distribution and specific surface area for the cementitious materials are given in Supporting Information 1, S1.1.

2.1.2. Sand and aggregates

All seven concretes contained the same quantity and type of sand and aggregate, further details available Table A2. Sand and aggregates were all sourced from Baywa. Particle size distribution for the sand and aggregates are given in Supporting Information 1, S1.2.

2.1.3. Chemical admixtures

Four commercial admixtures (two superplasticizers, one accelerator, and one water-repellant) were used. All admixtures were provided by Master Builders Solutions as water-based liquid solutions. The superplasticizer type for each concrete was selected based on compatibility with calcined clay in the binder (e.g., CEM II/A-LL vs LC³ mixes), as previous studies have shown calcined clay mineralogy impacts the performance of conventional PCE superplasticizers used in OPC concrete (Dhers et al., 2024). These were MasterGlenium SKY (solids content 25 g in 100 g solution, density 1.06 kg L⁻¹) for CEM II cement, and MasterCO₂re (based on clustered PCEs - solids content 25 g in 100 g solution, density 1.07 kg L⁻¹) for LC³ cement. The accelerator used was Master X-Seed (total solids content 22 g per 100 g solution, density 1.14 kg L⁻¹), which is a suspension of C-S-H nanoparticles designed to promote nucleation of hydration products at early-ages of hydration, to accelerate early-age strength development. The water repellent used was MasterPel (total solid content of 50 g solids per 100 g solution, density 1 kg L⁻¹), which is a silane-based suspension designed to improve water-resistance in concrete. Note that MasterGlenium, MasterCO₂re, Master X-Seed and MasterPel are registered trademark of Construction Research & Technology GmbH, a Master Builders Solutions company.

2.2. Mix design

All seven concretes contained superplasticizer. The superplasticizer dosage was adjusted (Section 2.4.1) until each mix achieved equivalent workability (50 ± 10 cm slump flow over 90 minutes). The only admixture added to the reference CEM II/A-LL was superplasticizer (MasterGlenium SKY), and none of the LC³ concrete mixes contained MasterGlenium SKY superplasticizer (this superplasticizer was not sufficient to achieve the required flow class when using LC³ binders). For the six LC³ concrete mixes (three LC³-50, three LC³-35), there were three levels of admixture addition: S (MasterCO₂re superplasticizer only); SA (MasterCO₂re superplasticizer and Master X-Seed accelerator); and SAW (MasterCO₂re superplasticizer, Master X-Seed accelerator, and MasterPel water repellent). The dosage of accelerator was doubled in LC³-35 mixes compared to LC³-50 mixes per manufacturer advice to compensate for the lower clinker content. The mix design for each concrete, including final dosages of admixtures, is given in Table 1. Additional detail regarding admixture dosing schedule is summarised in Table A3.

2.3. Concrete preparation

Concretes were prepared in the laboratory. For each mixture, three 100×100×100 mm cubes were cast: two for compressive strength testing and one for carbonation testing, due to a limited amount of calcined clay. Each specimen was removed from the moulds after 1 day, prior to curing in conditions required for compressive strength testing (Section 2.4.2) or accelerated carbonation (Section 2.4.3).

2.4. Mechanical and durability testing

2.4.1. Consistence/workability test

All seven concretes (Section 2.2) were tested for equivalent consistence based on the EN 12350-5 standard slump test (British Standards Institution, 2009). Fresh concrete was prepared (Section 2.3), and then filled into a cone with dimensions $r = 100$ mm, $R = 100$ mm, and $h = 100$ mm. Once filled, the fresh concrete was densified using a steel rod, and surplus material was removed by evening the surface. The cone was then lifted, and the slump measured using a ruler. After each measurement, the material was recollected and stirred immediately prior to the next measurement. Slump was measured at: 5, 15, 30, 60, and 90 minutes after initial mixing. The superplasticizer dosage was adjusted until all mixes achieved comparable slump over 90 minutes (50 ± 10 cm).

2.4.2. Compressive strength testing

Cubic concrete samples were first cured in water (1 – 7 days), and

then in 20 °C and 65% relative humidity air (7 – 28 days), per conditions specified in EN 12390-2 (British Standards Institution, 2019a). Compressive strength testing was carried out at 28 days per EN 12390-4 (British Standards Institution, 2023), using measured on a Zwick Roell Allround Line Z150 (ZwickRoell, Ulm, Germany). Compressive strength classes for cubic samples defined by EN 206 are given (Table A4).

2.4.3. Carbonation testing

Carbonation depth was measured using accelerated carbonation tests performed per EN 12390-10 (British Standards Institution, 2019b), in a carbonation chamber with 3% CO₂ at 20°C and 60% relative humidity. Measurements were taken at 28- and 70-days exposure, except for the CEM II/A-LL sample which was only measured at 70-days, by cutting a cross section in the sample and applying phenolphthalein solution and measuring the carbonation depth with a Vernier caliper gauge. The average value of 10 position measurements was taken per sample. Carbonation depth in mm was converted to carbonation rate using Eq. (1) (Papadakis et al., 1991b), using the carbonation depth measured at 70 days:

$$\text{carbonation rate} = \frac{\text{carbonation depth}_{\text{mm}}}{\sqrt{\text{days}}} \quad (1)$$

Carbonation durability class (XC1-4 per Table A5, and Tables S3-S4, Supporting Information 1, S4.0) for each concrete mix was determined based on the carbonation rate at 70 days and the resistivity values (Figure S3, Supporting Information 1, S3.0). The rate limits were selected per French standard XP P 18-458 (Association Francaise de Normalisation, 2013), which uses the XC1-4 carbonation environments per EN 206 (British Standards Institution, 2013).

2.5. Life cycle inventory

2.5.1. Goal and scope

The goal of this study is to analyse the impact of admixtures on the carbon footprint (as 100-year global warming potential, GWP) and technical performance of LC³ concretes with different clinker contents and admixture addition compared to CEM II/A-LL concrete. Technical performance refers to the 28-day compressive strength and carbonation rate for equivalent workability concretes. We consider a cradle-to-gate scope for each mix (e.g., per m³ concrete). We use mix volume (kg CO₂-eq. m⁻³ concrete) as the functional unit, and discuss the carbonation resistance and 28-day strength performance of each mix. Concretes are compared based on meeting the same minimum performance characteristics (e.g., minimum strength and carbonation resistance class), and we include data for concretes collected from the literature to

Table 1

Concrete mix design, given in kg component m⁻³ concrete. The dosage of admixtures (MasterGlenium SKY, MasterCO₂re, MasterPel, and Master X-Seed) are given in kg solution, including dilution factors per Table A3. Water is additional water, exclusive of water contained in admixtures. Total cement (kg cement m⁻³ concrete) and w/c ratio (mass water to mass of cement) added for information regarding mix design, but are not used in inventory calculations. Mix component GWP (kg CO₂-eq. kg⁻¹) used as inventory data summarised here for convenience, detailed in Section 2.5.2.

Mix:	GWP (kg CO ₂ -eq. kg ⁻¹)	CEM II/A-LL	LC3-50			LC3-35		
			S	SA	SAW	S	SA	SAW
Clinker	0.86	274.7	151.4	151.4	151.4	106.0	106.0	106.0
Limestone	5.4×10^{-3}	28.8	63.4	63.4	63.4	78.1	78.1	78.1
Gypsum	8.2×10^{-3}	16.6	16.1	16.1	16.1	16.7	16.7	16.7
Calcined clay	0.314	0.0	112.0	112.0	112.0	145.6	145.6	145.6
MasterGlenium SKY	0.57	0.7	0.0	0.0	0.0	0.0	0.0	0.0
MasterCO ₂ re	0.62	0.0	2.0	2.0	2.0	2.4	2.4	2.4
MasterPel	1.37	0.0	0.0	0.0	2.6	0.0	0.0	2.6
Master X-Seed	0.73	0.0	0.0	1.5	1.5	0.0	3.0	3.0
Total sand	5.4×10^{-3}	1002.4	1002.4	1002.4	1002.4	1002.4	1002.4	1002.4
Total aggregate	5.4×10^{-3}	928.9	928.9	928.9	928.9	928.9	928.9	928.9
Water	0.00	148.4	148.4	143.1	140.5	141.6	130.9	128.3
Total cement		320	343	343	343	346	346	346
w/c ratio		0.47	0.45	0.45	0.45	0.43	0.43	0.43

increase comparable points (Section 2.6). We did not consider any additional transport emissions (e.g., between sites and concrete batching facility) as we assumed all production occurred in the same region.

2.5.2. Inventory analysis

The foreground systems (e.g., each concrete mix) considered the amount of each component used per m^3 of concrete (Table 1, Section 2.2). Table 1 summarises the data used for inventory analysis, with detailed information available in Table A6. Where available, data was taken from ecoinvent database v3.9.1 (Swiss Centre for Life Cycle Inventories, 2022). This includes the data for clinker, limestone, and gypsum, and their carbon footprint (GWP, $\text{kg CO}_2\text{-eq. per kg material}$) values were calculated using the 'EF v3.1 EN15804 climate change (GWP100)' impact assessment method for consistency with the EPD data used for admixtures. Data for cradle-to-gate production of calcined clay (including quarry extraction, calcination, and grinding) were taken from (Miller et al., 2018) and calculated using 'EF v3.1 EN15804 climate change (GWP100)' using background data from ecoinvent 3.9.1 for European production and coal-fired heat, as these data have been reviewed to be the most representative inventory available of calcined clay production compared to the simplified dataset available in ecoinvent 3.9.1 (Edwards et al., 2026b).

No detailed process inventory is available for all admixtures used in this study due to confidentiality concerns, therefore data for each admixture were provided by Master Builders Solutions from each product's Environmental Product Declaration (EPD) (total $\text{CO}_2\text{-eq.}$, A1-A3 category to align with cradle-to-gate scope, total $\text{CO}_2\text{-eq. per kg material}$). To validate the EPD figures, we compared against previously reported inventory for PCE based superplasticizers, 2 – 2.6 $\text{kg CO}_2\text{-eq. kg}^{-1}$ product on a dry basis (Schiefer and Plank, 2023b). The PCE based superplasticizers used in this study are liquid products (water-based solution) with a 25% solids content (per Table A3 in Appendix), for which the A1 – A3 reported EPD results (0.57 and 0.62 $\text{kg CO}_2\text{-eq. kg}^{-1}$ for both PCE products used per Table A6 in Appendix) are within the reported range of results on a dry basis. Therefore, we considered the EPD data provided by the manufacturer a suitable estimation of the cradle-to-gate production inventory.

2.6. Literature data

Data were collected from the literature (i.e., (Bucher et al., 2017; Khan et al., 2018; Kim et al., 2007; Rutkowska et al., 2023; Zhao et al., 2026)) where concrete mix designs, 28-day compressive strengths, and carbonation rates were all specified, resulting in a dataset including 35 concrete samples. The binders used in these concretes included cement with 4.5 – 90 wt.% clinker content, and incorporated clinker substitutes such as granulated blast furnace slag (GBFS), fly ash, silica fume, and calcined clay. The 28-day strength results of these concretes range from 35.5 – 79 MPa, and carbonation rates ranged from 0.3 – 4.7 $\text{mm d}^{-0.5}$. The clinker content in cement, calculated carbon footprint (per Section 2.5), 28-day strength, carbonation rate, and resulting strength and carbonation classes (per Section 2.4.2 and 2.4.3) of these concretes are given in Supporting Information 2, Tab 1.

3. Results and discussion

This study investigated the impact of different admixtures (superplasticizer, accelerator, and water repellent) on the workability, strength, and carbonation resistance of concretes made with CEM II/A-LL, LC³-50, and LC³-35 cements. We find that the use of admixtures with LC³ can result in concrete with performance comparable to that of CEM II/A-LL, while reducing carbon footprint by 22 – 38% per m^3 . We note that these LC³ concretes would not have been able to reach similar performances in workability, particularly slump retention, and compressive strength, without the use of the superplasticizer and accelerator.

3.1. Workability

Superplasticizer dosage was optimised for each mix such that all concretes had equivalent workability (50 ± 10 cm slump flow) over 90 minutes. This slump flow is equivalent to F4 – F5 flowability class per EN 206 ((British Standards Institution, 2013)), and is suitable for applications such as ready mix concrete. The final dosage of superplasticizer in each blend enabled comparable workability between concretes with different water to cement ratios: 0.47 (CEM II/A-LL), to 0.45 (LC³-50 mixes), and 0.43 (LC³-35 mixes). Reducing the w/c ratio generally increases the strength and durability of concrete by creating a denser matrix structure (Kim et al., 2014), and therefore adding superplasticizer to LC³-50 and LC³-35 concretes should benefit strength performance and carbonation resistance.

For LC³-50 and LC³-35 concretes, initial slump flow (5 – 30 minutes) was highest in the concretes containing accelerator, water repellent and superplasticizer. The initial (5 – 30 minutes) workability was lowest in LC³-50 and LC³-35 concretes containing only superplasticizer, however these concretes had the highest workability from 60 – 90 minutes compared to the LC³-35 and LC³-50 concretes with other admixtures. This is likely due to the delayed release of the superplasticizer polymers used in the LC³ blends (MasterCO₂re) and suggests the water repellent (MasterPel) improves the initial workability of LC³ concretes prior to the release of the superplasticizer polymers. Workability measurements are available in Fig. A1.

3.2. Compressive strength

Compressive strength of the concretes was measured at 1 and 28 days of curing (Fig. 1). CEM II/A-LL concrete had the highest overall 1-day compressive strength (16.8 MPa). LC³-50-SA (e.g., containing superplasticizer and accelerator) had the highest 1-day compressive strength performance of the LC³ mixes (82% of CEM II/A-LL), and LC³-35-SA had the highest 1-day compressive strength performance of the LC³-35 mixes (65% of the CEM II/A-LL compressive strength).

Therefore, of the admixtures, the accelerator had the greatest effect on 1-day compressive strength for LC³ concretes. The accelerator improved the early strength performance by 15% for LC³-50 concretes (comparing LC³-50-SA to LC³-50-S), and 50% for LC³-35 concretes (comparing LC³-35-SA to LC³-35-S). Overall, these results are consistent with 1-day compressive strength results for LC³-50 mortars using same accelerator (e.g., C-S-H nanoparticle accelerator), which achieved 1-day compressive strengths of 22 – 29 MPa in mortars with a w/c ratio of 0.4 and target a workability of 20 ± 1.5 cm flow (Morales-Cantero et al., 2023). However, this previous study used a higher dosing rate of accelerator (2 wt.% by weight of binder, compared to a maximum of 0.43 wt.% by weight of binder for LC³-50 used in this work) and the resulting mortar was significantly less workable than the results presented here. This demonstrates the design trade-off between different parameters (e.g., workability vs. early strength) and the range of performance that can be achieved with incorporating admixtures in concrete mix design.

All LC³ mixes, excluding LC³-35-S, had up to 21% higher 28-day compressive strength than CEM II/A-LL, and achieved at least 60 MPa compressive strength at 28 days of curing (C50/60 strength class concrete). Without additional admixtures (e.g., accelerator and water repellent), the LC³-35-S compressive strength performance was insufficient for C50/60 class strength performance equivalent to the CEM II/A-LL. Adding accelerator increased the 28-day compressive strength of LC³-35-SA by 31% compared to LC³-35-S at 28-days curing, which was the largest step change in 28-day compressive strength performance of any mix, and enabled comparable LC³-35 compressive strength performance to CEM II/A-LL concrete at 28 days curing. This is consistent with previous results for LC³, as LC³-50 without admixtures can have equivalent or higher 28-day compressive strength than OPC (e.g., (Avet and Scrivener, 2018; Scrivener et al., 2018c; Zunino and

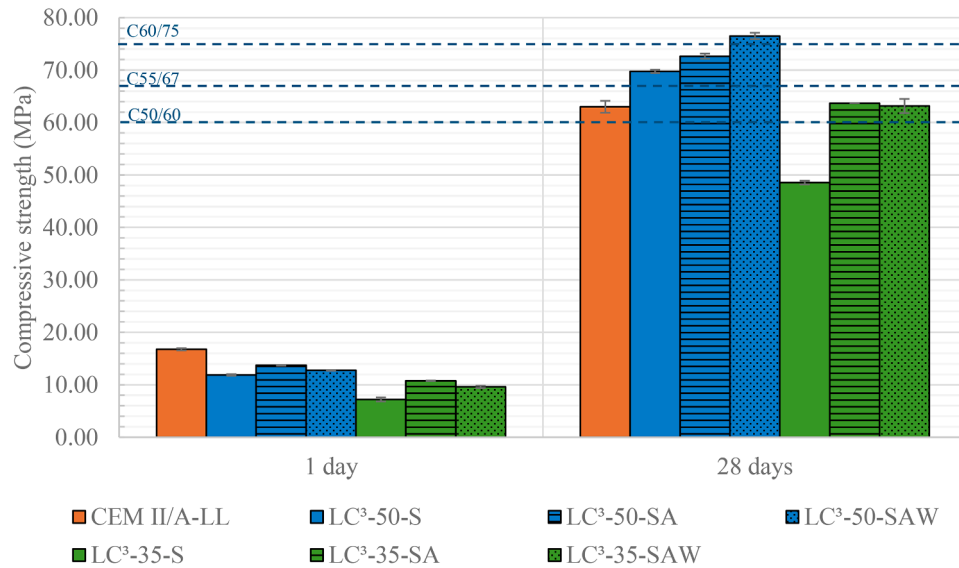


Fig. 1. Average compressive strength (MPa) test results at 1- and 28-days for the seven concretes: CEM II/A-LL (orange), LC³-50 (blue), and LC³-35 (green). Unpatterned series contain only superplasticizer, horizontally striped series contain superplasticizer and accelerator (Master X-Seed), and dotted series contain superplasticizer, accelerator (Master X-Seed) and durability enhancing admixture (MasterPel). Error bars represent one standard deviation. Horizontal dotted lines represent the minimum 28-day compressive strength for C50/60 (min 60 MPa), C55/67 (min 67 MPa) and C60/75 (min 75 MPa) compressive strength classification.

Scrivener, 2025)), whereas LC³-35 cement without admixtures (e.g., LC³-35-S) have lower 28-day compressive strength than OPC (Sun et al., 2024; Zunino and Scrivener, 2025). While the present study focuses on 1 and 28 day strength, previous studies have also demonstrated that LC³ systems continue to develop strength past 28 days due to ongoing pozzolanic reaction and contribution of carboaluminate phases (Sun et al., 2024). Therefore, further reduction in clinker factor is possible for applications which do not require high early-age strength.

LC³-50-SAW (e.g., containing superplasticizer, accelerator, and water repellent) had the highest 28-day compressive strength of all samples (76.5 MPa), 21% higher than that of CEM II/A-LL and 5% higher than that of LC³-50-SA. This is in contrast both to the result for LC³-35-SAW, which had 0.8% lower 28-day strength performance compared to LC³-35-SA, and to reported results for OPC concretes where the addition of silane-based water repellent reduced 28-day compressive strength (Hossain et al., 2024; Zhang et al., 2022; Zhu et al., 2013).

However, to the authors' knowledge, there are no reported results for silane addition on the compressive strength performance of LC³ concretes. Therefore, the impact of silane water repellents on the strength of LC³-50 and LC³-35 concretes should be further tested with and without accelerators to validate the result presented here.

3.3. Carbonation

The carbonation depth was measured at both 28 and 70 days of curing (Fig. 2). CEM II/A-LL had the lowest carbonation depth at 70-days, meaning that this concrete had the highest carbonation resistance of all mixes. Carbonation depth increased with decreasing clinker content, which is consistent with previous studies (e.g., (Scrivener et al., 2018c; Bucher et al., 2015; Khan et al., 2018; Bucher et al., 2017)).

LC³-35-S had the highest carbonation depth at both 28 and 70 days (18.3 mm and 26.8 mm respectively). At 70 days both LC³-35-SA (20.3

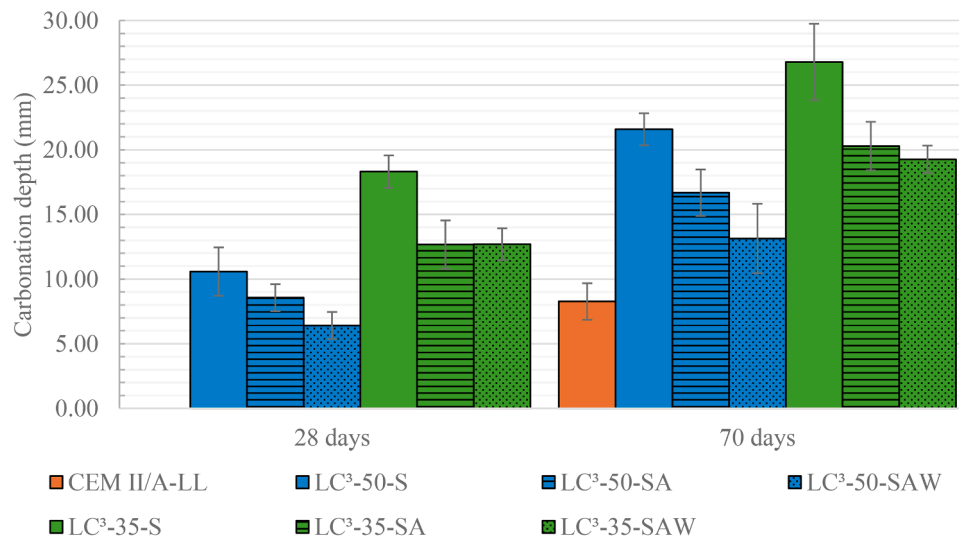


Fig. 2. Average depth of carbonation front (mm) at 28 days and 70 days for each sample: CEM II/A-LL (orange), LC³-50 (blue), and LC³-35 (green). Unpatterned series contain only superplasticizer, horizontally striped series contain superplasticizer (MasterCO₂re) and accelerator (Master X-Seed), and dotted series contain superplasticizer (MasterCO₂re), accelerator (Master X-Seed) and durability enhancing admixture (MasterPel). Error bars represent one standard deviation.

mm) and LC³-35-SAW (19.3 mm) recorded lower carbonation depths than LC³-50-S (21.6 mm), demonstrating that adding admixtures to LC³-35 concrete can achieve comparable carbonation performance to LC³-50 concrete without admixtures. The impact of water repellent was greater on LC³-50 mixes than LC³-35 mixes, with LC³-50-SAW having a 21% lower carbonation depth than LC³-50-SA, compared to LC³-35-SAW having a 5% lower carbonation depth than LC³-35-SA. However, the dosing rate of accelerator (Master X-Seed) was higher in LC³-35 mixes than in LC³-50 mixes (Table 1), while water repellent (MasterPel) dosage was the same for both LC³-35 and LC³-50. This suggests that increasing accelerator dosage improves carbonation resistance for LC³ concretes, and dosing rate of water repellent may need to increase at high accelerator dosages to achieve significant effects. The carbonation rates (per Eq. (1)) at 70 days and carbonation resistance classes (Section 2.4.3) are shown against 28-day compressive strength in Fig. 4. CEM II/A-LL and LC³-50-SAW had the lowest carbonation rates (1 mm d^{-0.5} and 1.55 mm d^{-0.5} respectively), both suitable for 100-year lifetime under XC3 carbonation exposure conditions (Section 2.4.3). CEM II/A-LL had the lowest overall carbonation rate with a value (1 mm d^{-0.5}) much below the 100-year XC3 carbonation resistance threshold (1.8 mm d^{-0.5}, Section 2.4.3), indicating that CEM II/A-LL is oversized.

Adding accelerator to LC³-35 concretes (e.g., LC³-35-SA and LC³-35-SAW) reduced their carbonation rate to below that of LC³-50-S. Additionally, admixtures widened the suitable carbonation exposure environments for LC³-35 concretes: LC³-35-SA and LC³-35-SAW were suitable for 100-year lifetime under XC2 carbonation exposure class, whereas LC³-35-S was only suitable for XC1. However, there was no significant change in carbonation rate or strength between LC³-35-SA and LC³-35-SAW, indicating limited impact of water repellent on LC³-35 already dosed with accelerator. LC³-50-SA had higher carbonation resistance class than LC³-50-S, sufficient for 100-year lifetime under XC4 carbonation exposure conditions, but not XC3, for its C55/67 strength class concrete. Both LC³-50-SAW and CEM II/A-LL were suitable for 100-year lifetime under the most aggressive carbonation exposure conditions (XC3), with CEM II/A-LL having a lower strength class (C50/60) but a higher margin to the carbonation rate specification limit for XC3 compared with LC³-50-SAW. LC³-50-SAW had a higher strength class

(C60/75), but smaller margin to the XC3 carbonation rate specification limit.

In general, Fig. 3 shows that higher strength LC³ concretes had higher carbonation resistance (i.e., lower carbonation rate). This relationship is considered to be both true and an oversimplification (Neville, 2011), and is better explained by the effect of the C-S-H nanoparticle accelerator on the concrete microstructure. Thomas et al. (2009) demonstrated that adding C-S-H nanoparticle based accelerators to concrete produces a more uniform microstructure with lower porosity by promoting precipitation of hydrates in the capillary pores (Thomas et al., 2009). Lower porosity generally reduces diffusivity through the concrete matrix, thus limiting its effective carbonation rate. Overall, these results indicate that accelerator is more impactful to concrete strength and carbonation class than water repellent, except for LC³-50-SAW. The effects of water repellent should be further studied in harsh carbonation environments (e.g., in the case of LC³-50-SAW).

3.4. Life cycle assessment

Fig. 4 shows the carbon footprint (kg CO₂-eq. m⁻³ concrete) of each mix as total values and disaggregated by component material. The CEM II/A-LL mix had the highest total carbon footprint value of 249.5 kg CO₂-eq. per m³ concrete. All LC³ mixes (154 – 194 kg CO₂-eq. m⁻³ concrete) reduced carbon footprint compared with CEM II/A-LL due to their reduced clinker content. LC³-50 has 22–27% lower carbon footprint compared to CEM II/A-LL, whereas LC³-35 has 31–38% lower carbon footprint compared to CEM II/A-LL, with the range in results depending on admixture content. Over 50% of total CO₂-eq. emissions of each mix were from cement clinker (e.g., from 53% of LC³-35-SAW to 95% of CEM II/A-LL), making it the largest source of emissions in each concrete. Calcined clay was the second largest source of emissions for all LC³ concretes (18 – 30% of their total emissions), demonstrating the potential of calcined clay decarbonisation in further reducing the carbon footprint of LC³ concretes.

In this study, we have used a value of 0.31 kg CO₂-eq. per kg calcined clay that in our opinion reflects a conservative estimate of the carbon footprint of European calcined clay production using thermally driven

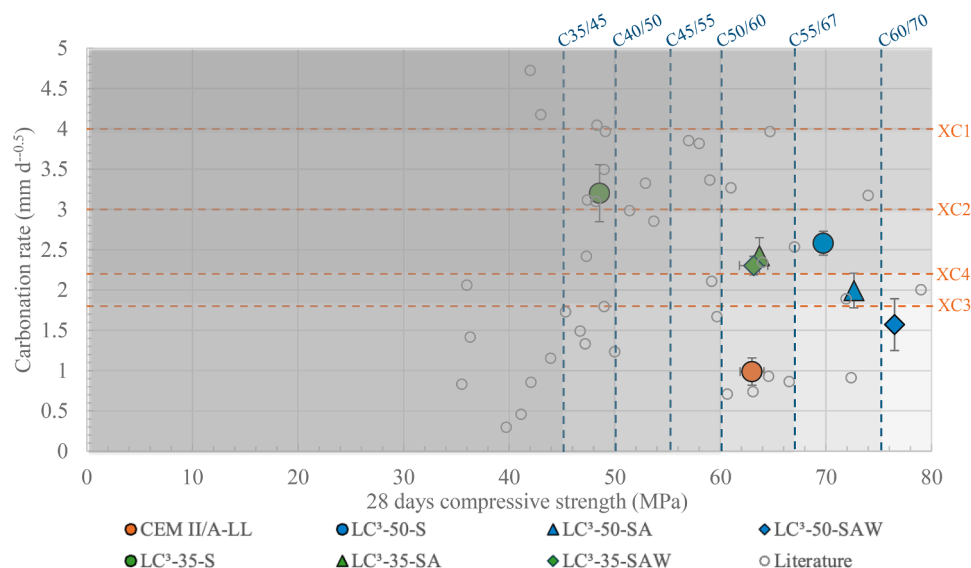


Fig. 3. Carbonation rate vs 28-day compressive strength of each concrete: CEM II/A-LL (orange), LC³-50 (blue) and LC³-35 (green). Different superplasticizer addition to LC3 mixes is differentiated by dot shape: superplasticizer only ('S' mix, circle), superplasticizer and accelerator ('SA' mix, triangle), and superplasticizer, accelerator and water repellent ('SAW' mix, diamond). Horizontal orange dotted lines are the threshold (maximum) carbonation rates for 100-year structure lifetime under carbonation exposure per Table A5, and each classes applicable carbonation rates are shaded in overlapping grey (e.g., lighter shading = higher specification). Vertical blue dotted lines show the minimum 28 days compressive strength for cubic concrete sample classification per Table A4, and each classes applicable strength are shaded in overlapping grey (e.g., lighter shading = higher specification), with classes labelled vertically above the chart (strength), and horizontally to the right of the chart (carbonation resistance) against the corresponding dotted lines. Unfilled grey circles represent literature data collected (Section 2.6).

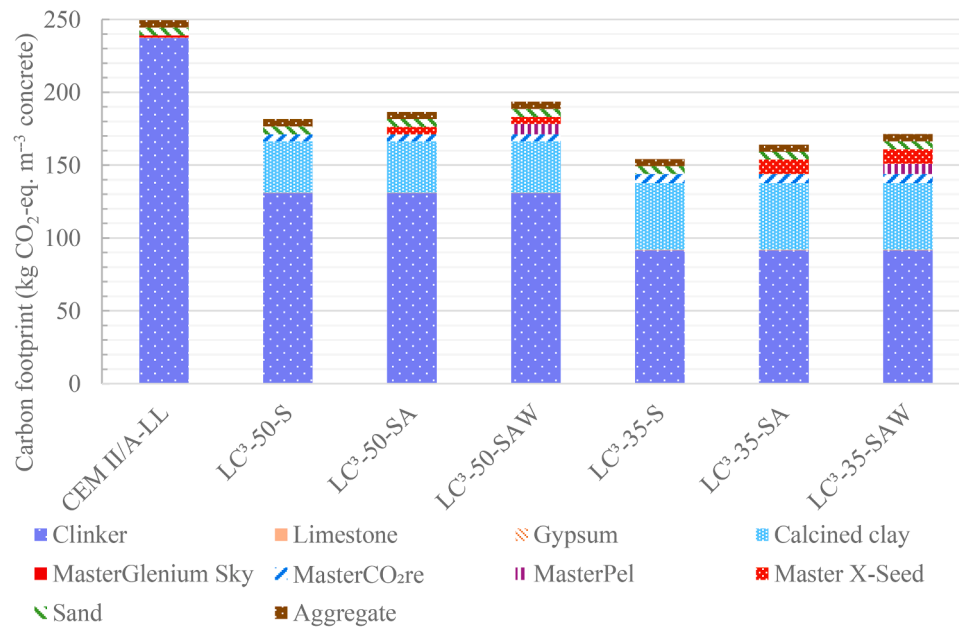


Fig. 4. Total carbon footprint for each concrete mix showing the contribution of each component in the concrete (water not shown as contribution was not visible due to negligible contribution to the total carbon footprint in each concrete).

calcination processes. We also considered a scenario where the calcined clay production had a carbon footprint of 0.17 kg CO₂-eq. per kg calcined clay, which is reported by French producers with access to decarbonised electricity from nuclear energy and use of biogas from agricultural waste (Argenco Development, 2024; Habert and Ouellet-Plamondon, 2016). While this figure for calcined clay demonstrates the further decarbonisation potential of calcined clay production, and reduces the carbon footprint of the LC³ mixes studied here by 29–46 % (133–170 kg CO₂-eq. per m⁻³ concrete) compared to the CEM II/A-LL, this lower impact calcined clay is still the second-largest single contribution to the carbon footprint of the mix (refer to Supporting Information 1, S5.0).

While each concrete mix contains a relatively low (e.g., <1 wt.%) mass of admixtures, admixtures contributed between 0.6% of the CO₂-eq. emissions for CEM II/A-LL (which only contained the lowest impact admixture, MasterGlenium SKY), and 13.5% for LC³-35-SAW (which contained all three admixtures at the highest dosing rates). Admixture contribution is more impactful as clinker content decreases, particularly where multiple admixtures are used, but still significantly lower than that of clinker (13.5% vs 53% in LC³-35-SAW) and the total binder (13% vs 80% in LC³-35-SAW). We note the use of EPD data to fill inventory data gaps for the admixtures, which is an approach under active investigation (e.g., (Del Rosario et al., 2021; Galindro et al., 2020; Schindler et al., 2025; Strazza et al., 2016)). However, compared to process-based inventories, reported impact results in EPDs are a ‘black-box’, and it is difficult to verify the ISO 14044 requirements (e.g., for consistency, system boundary coherence, allocation alignment), and further analysis of the environmental impact of admixture production (e.g., hotspot analysis to identify decarbonisation opportunities within admixture production) cannot be performed. Better understanding of the environmental impacts of admixture production from producing transparent inventory data for admixtures is therefore an important enabler for further decarbonising admixture production. However, the largest potential for reducing cement global warming potential is clearly from reducing the clinker factor in cement. We recommend future work produces a detailed inventory of C-S-H nanoparticle accelerator and silane-based water repellent (whether based on operational production data in collaboration with producers, or independently via process simulation and other prospective LCA techniques (Tsoy et al., 2020)),

and identify the area of EPD result integration in process-based LCA as a topic for future methodological research in industrial ecology.

3.5. Performance summary and insights

Fig. 5 shows carbon footprint (as GWP) vs clinker content in binder for concretes in strength performance class C35/45, C50/60, C55/67, and C60/75 and 100-year lifetime for carbonation exposure conditions XC1-4.

Low clinker LC³ (LC³-50 and LC³-35) concretes had the lowest carbon footprint per m⁻³ concrete for each strength and carbonation exposure class. For C50/60 XC2 or lower specification classes, the optimum concrete was an LC³-35, while for higher specification classes the optimum concrete was an LC³-50. Generally, the best mix contained superplasticizer and accelerator, except for C35/45 XC1 (where the optimum concrete was LC³-35-S with superplasticizer only), C55/67 XC1-2 (where the optimum concrete was LC³-50-S with superplasticizer only), and all C35/45-C60/75 XC3 carbonation exposure conditions (where the optimum concrete was LC³-50-SAW containing superplasticizer, accelerator, and water repellent). Therefore, admixtures should be considered a priority for expanding the suitable applications of low clinker content cement and concretes.

Although this study focusses on carbon footprint as the primary sustainability indicator for the assessed binders, producing LC³-based concretes and structures also improves performance compared to OPC binder systems in other environmental, social and economic dimensions (UNEP/SETAC Life Cycle Initiative, 2011). For example, calcined clay can be produced from widely available common clays which means local production is possible in many regions (Scrivener et al., 2018c), improving accessibility compared to high-clinker binders, as clinker must be imported in regions with poor availability of the high-quality limestone. Additionally, reducing clinker content in cement extends the reserves of these high-quality limestones, and potentially reduces land-use per kg of binder as fewer mineral resources are converted to volatile emissions at high temperature (Edwards et al., 2026a). Future work could also explore the trade-off between conflicting criteria (e.g., environmental and technical performance) using multi-criteria decision making tools, which can remove subjectivity between criteria (Sánchez-Garrido et al., 2022).

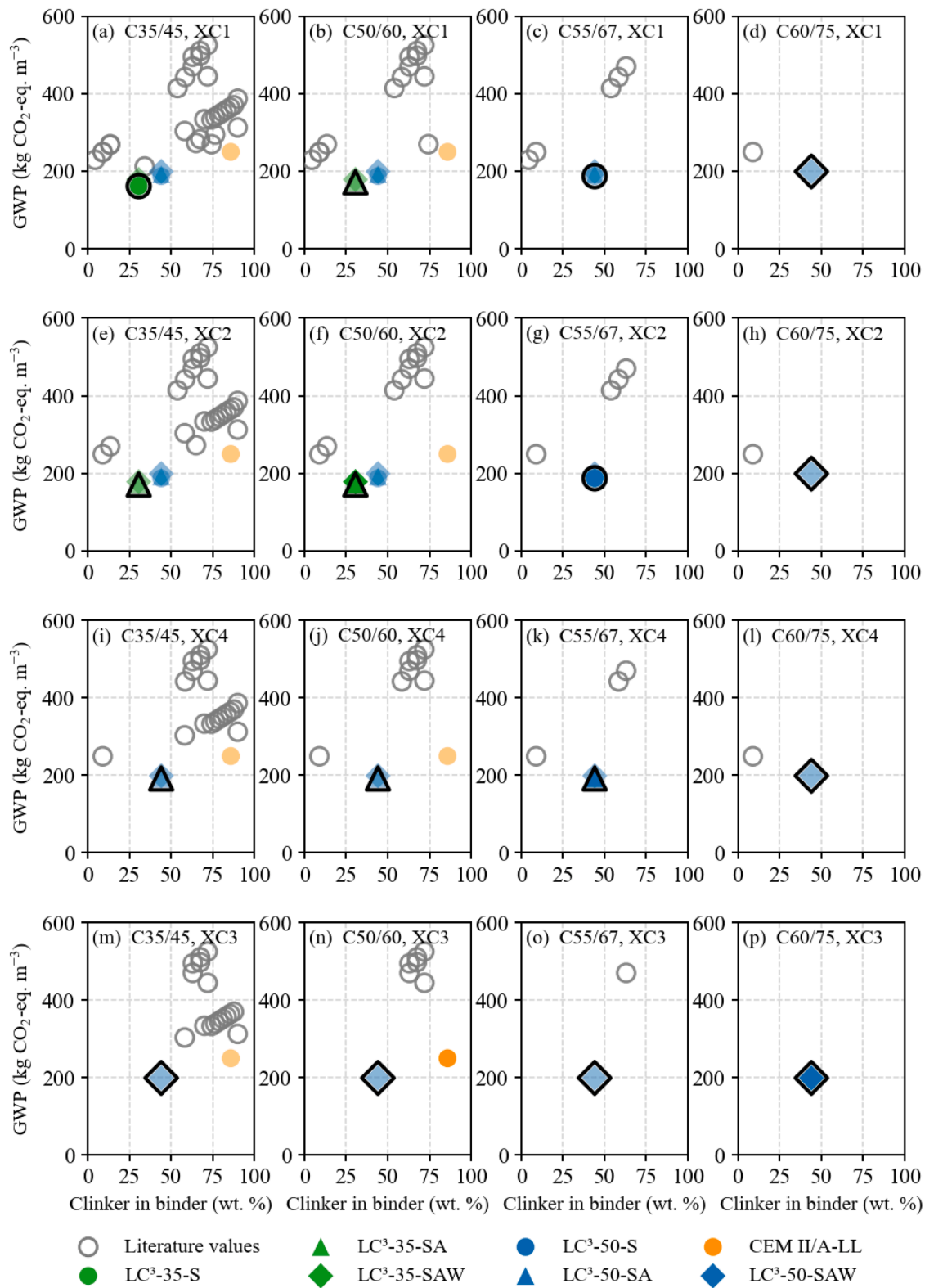


Fig. 5. Concrete carbon footprint (given as GWP, kg CO₂-eq. m⁻³ concrete) vs clinker content in binder for concretes meeting the same minimum strength and carbonation exposure classes. The black outlined point is the concrete with the lowest impact per m⁻³ in the subplot. Opaque data points indicate the highest specification class for each concrete mix (e.g., (p) C60/75 XC3 concrete is the highest specification achieved by the LC³-50-SAW mix). Partially transparent filled points indicate concrete mixes in each plot class which also meet higher specification classes (strength and/or carbonation exposure). Unfilled grey circles represent literature data collected (Section 2.6).

We derive the following heuristics to enable practical addition of accelerator and water repellent admixtures to minimise the overall carbon footprint of LC³-based concrete and structures. Firstly, our results (Figs. 3 and 5) show that admixtures enable LC³ concretes be applied to higher specification classes. In general, adding C-S-H seed accelerators to concretes, including LC³, generally improves strength performance sufficient for an increase in strength class (He et al., 2021; Li et al., 2024; Morales-Cantero et al., 2023; Vallina et al., 2026). This adds to the literature showing that clinker content can be reduced if a C-S-H seeds accelerator is added to maintain the same strength performance. The findings of this work should be applied to other composite cements (e.g., those containing limestone and natural pozzolans, which are less reactive than calcined clay (Edwards et al., 2026a)), to determine if the performance enhancements from superplasticizer and accelerator can maintain concrete compressive strength performance at lower clinker contents, and thus support wider use of these materials.

Secondly, we found that accelerator addition was effective at decreasing the carbonation rate of concrete in addition to increasing strength (Fig. 3), and generally accelerator addition also improved carbonation exposure class (e.g., XC1 → XC2 for LC³-35 and XC2 → XC4 for LC³-50). However, both accelerator and water repellent admixture were required for LC³-50 concrete in the most severe carbonation risk environment (XC3), where very-high carbonation resistance is required. Therefore, concrete mix design should first consider maximising the amount of accelerator prior to adding water repellent, as accelerator has the lower carbon footprint (Table A6) and contributes to enhancing both the strength performance and carbonation resistance. Water repellent can then be considered as a further addition for concretes in severe carbonation exposure conditions. Alternatively, other mitigations for preventing carbonation corrosion of reinforced steel concretes (e.g., applying coatings on the concrete, surface modification of rebar, increased concrete maintenance and repair strategies) could be considered for the highest-risk environments, and the cradle-to-gate carbon footprint of these techniques against the use of water-repellent admixtures should be explored in future LCA studies. Additionally, reducing carbonation rate in concretes to extend durability should be explored against the potential for carbonation to sequester CO₂ during its in-use and end-of-life phase, and the suitability of highly carbonated concrete to be recycled into reactive binder, which would require future studies to consider a cradle-to-grave boundary.

Finally, higher specification concretes could be the optimum choice for lower class applications (e.g., LC³-50-SAW was optimum for C35/45 – C50/60 XC3 conditions, despite this concrete having C60/75 class strength), depending on the concrete mixes available. Additionally, use of higher specification concretes could enable the overall concrete volume (or cement content in concrete) to be reduced in the applications with lower specification requirements. Future work should also assess the possible reduction in concrete volume possible from higher specification concrete in lower specification applications, to access the low strength-weighted emissions available. Fig. 5(a-l) show that LC³-35 and LC³-50 concretes outperformed concretes with very low (<25 wt.%) clinker in binder content. These very low clinker in binder concretes were made with materials including GBFS and a phosphogypsum material (Zhao et al., 2026), however had higher amounts of binder (680 kg per m³ concrete) than the LC³ concretes investigated in this study (320 – 346 kg cement per m³ concrete per Table 1). Therefore, both the clinker content, and the total binder requirement are key variables for analysing the carbon footprint of concrete.

3.6. Discussion on potential mechanism of action

During carbonation, CO₂ diffuses through the pore network of partially saturated cement paste and dissolves into the pore solution, where it reacts with the Ca-bearing hydrate phases (predominantly portlandite and C-S-H) to form CaCO₃ (Georget et al., 2018). The rate of carbonation is generally observed to follow a $\sqrt{t}$ relationship, suggesting

the CO₂ diffusion is the rate-limiting step, however the rate is also influenced by pore structure, moisture state, concentration of CO₂ and the availability of Ca-bearing hydrates in the pore solution (Huet et al., 2011). One challenge with low-clinker cements is the modified hydrate chemistry, where less portlandite is typically available in the pore solution to buffer against CO₂ (von Greve-Dierfeld et al., 2020). Recent studies on low-clinker binders demonstrates that clinker factor can affect both the pore solution chemistry and the transport properties (e.g., microstructure) simultaneously (Dudi and Bishnoi, 2026, 2025; Singh et al., 2025). Experimental work (e.g., (Boumaaza et al., 2020)) and microstructure-based transport models (e.g., those by Georget, Prévost & Huet (Georget et al., 2018)) have demonstrated the relationship between pore structure parameters such as mean pore diameter and connectivity and gas diffusivity, reinforcing the dependence of carbonation rate on the diffusion of CO₂ through the pore network.

C-S-H seeding accelerators tend to reduce capillary porosity and refine the pore size distribution at early ages by promoting earlier nucleation of C-S-H and accelerating hydration reactions (John et al., 2018). In addition to promoting early-age strength development through accelerated hydration and denser microstructure (Sun et al., 2017), this refinement effectively reduces the CO₂ diffusion coefficient, limiting ingress of CO₂ into the pore solution, and thus limiting the propagation of the carbonation front (Boumaaza et al., 2020). The performance of C-S-H seeding accelerators in low-clinker cements is well established, having been validated in cements containing slag (Li et al., 2022), fly-ash and calcined clay (Cuesta et al., 2023; Morales-Cantero et al., 2024; Vallina et al., 2026). C-S-H seed accelerators are therefore an important tool to compensate for the lower buffering capacity in low-clinker cements, by improving the early hydration kinetics and refining the pore structure to limit CO₂ diffusivity and slow the transport of the carbonation front.

Consequently, combining low-clinker formulations with C-S-H seeding technologies, as shown in our study, may help balance sustainability objectives with durability requirements by simultaneously optimizing hydrate formation, pore structure, and carbonation resistance, which should be further investigated using detailed microstructural analysis to confirm the proposed mechanisms of action and identify further optimisation opportunities both in admixture design and concrete mix formulations. Such an in-depth microstructural investigation, using techniques such as SEM, XRD, or MIP, was beyond the scope of the present study and would constitute a dedicated research project; it therefore represents a logical next step for future work.

4. Conclusions

This study evaluated the impact of superplasticizer, accelerator, and water repellent admixtures on the fresh, hardened, and environmental performance of LC³ concretes. Both LC³-50 (50% clinker) and LC³-35 (35% clinker) cements were studied against CEM II/A-LL cement in concretes. We found that the use of admixtures improved the strength and carbonation resistance of LC³-50 and LC³-35 cement, while reducing the carbon footprint. In general, higher strength concretes had higher carbonation resistance, and we found that accelerator addition had the most strength performance and carbonation resistance benefit for LC³ concretes. While LC³-50 concrete with both accelerator and water repellent admixture was found to be the best performing mix, with the highest 28-day strength and very-high carbonation resistance, our LCA study demonstrated that the optimum concrete mix depended on the required strength class and carbonation exposure conditions of its intended application. Admixtures have been so far neglected in LCAs of concrete, however this work shows that admixtures contribute non-negligibly to the CO₂-eq. cradle-to-gate emissions of LC³-50 and LC³-35 concretes. This result demonstrates that should be included in the scope of LCA studies of low clinker concretes. Overall, we find that admixture addition enhances the performance of LC³ concretes, and enables high performance, low carbon concrete production.

CRediT authorship contribution statement

Pippa Edwards: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Sébastien Dhers:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization. **Rupert J. Myers:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

Sébastien Dhers is employed by Master Builders Solutions. This study was funded by Master Builders Solutions and materials and data were provided by Master Builders Solutions, however PE and RJM declare that this work represents their own objective and impartial analyses of

those data provided.

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Supplementary materials

Supporting material associated with this article can be found, in the online version, at [doi:10.1016/j.resconrec.2026.109095](https://doi.org/10.1016/j.resconrec.2026.109095). Supporting Information 1 (word document) contains particle size distribution for concrete materials and supporting data for carbonation results. The literature data used, including concrete class, GWP, clinker content and binder amounts, are available in Supporting Information 2 (as a Microsoft Excel file).

Appendix

The following tables and figures were moved to the Appendix to comply with journal requirements.

Table A1

Mineral composition of cementitious materials (wt.%); CEM II 42.5R (used as CEM II/A-LL cement) and CEM I 52.5R (used in LC3 mixes), limestone, calcined clay, and gypsum.

Phase [wt.%]	CEM II	CEM I	Limestone	Calcined clay	Gypsum
C ₃ S	55.2	59.7			
C ₂ S	13.4	13.7			
C ₃ A	8.3	8.8			
C ₄ (A,F)	6.7	6.4			
CaSO ₄	2.3	2.7			2.9
CaSO ₄ ·0.5H ₂ O	1.1	2.3			
CaSO ₄ ·2H ₂ O	1	0.2			97.1
K ₂ SO ₄	0.6	0.2			
K ₃ Na(SO ₄) ₂	0.1	0.2			
K ₂ Ca(SO ₄) ₂ ·H ₂ O	0	0			
K ₂ Ca ₂ (SO ₄) ₃	0	0			
CaO	0.3	0.3			
MgO	0.4	0.3			
CaCO ₃	8.3	3.2	98.7	0.6	
CaMg(CO ₃) ₂	0.6	1.2		0.2	
SiO ₂	0.7	0.6	1.3	47.3	
Ca(OH) ₂	0.7	0.3			
3Al ₂ O ₃ ·2SiO ₂				3.8	
Fe ₂ O ₃				0.7	
TiO ₂				1.7	
Al ₄ (OH) ₈ ·Si ₄ O ₁₀				2.1	
Amorphous				43.6	

Table A2

Sand and aggregate types and quantities used in each concrete mix.

Type	Size range (mm)	Quantity (kg m ⁻³ concrete)
Sand	0.1 - 0.3	9.7
Sand	0.3 - 1.0	102.9
Sand	0 - 4	889.8
Aggregates	2 - 5	38.2
Aggregates	5 - 8	424.3
Aggregates	8 - 11	221.4
Aggregates	11 - 16	244.9

Table A3

Summary of admixtures added to each concrete mix, with solid content (wt.% total solids in solution, per [Section 2.1.3](#)) and density. Code refers to the letter assigned to each of MasterCO₂re, Master X-Seed and MasterPel used as a suffix to differentiate the different LC³ mix designs (e.g., LC³-50-SAW is an LC³-50 mix that contains superplasticizer (S), accelerator (A), and water repellent (W)). ‘X’ denotes that the admixture was added to that mix.

	MasterGlenium SKY	MasterCO ₂ re	Master X-Seed	MasterPel
Admixture type	Superplasticizer	Superplasticizer	Accelerator	Water repellent
Code	N/A	S	A	W
Total solids (wt.%)	25%	25%	22%	50%
Density (kg m ⁻³)	1.06	1.07	1	1.14
Mix name:				
CEM II/A-LL	X			
LC3-50-S		X		
LC3-50-SA		X	X	
LC3-50-SAW		X	X	X
LC3-35-S		X		
LC3-35-SA		X	X	
LC3-35-SAW		X	X	X

Table A4

Characteristic concrete cube compressive strength per strength class at 28 days curing per EN 206 ([British Standards Institution, 2013](#)).

Class	Minimum cube strength (MPa at 28 days)
C35/45	45
C40/50	50
C45/55	55
C50/60	60
C55/67	67
C60/75	75

Table A5

Threshold carbonation rates (mm √d⁻¹) for carbonation exposure classes per XP P 18-458 French standard ([Association Francaise de Normalisation, 2013](#)), for 50-year and 100-year structure lifetime.

	50-year (mm √d ⁻¹)	100-year (mm √d ⁻¹)	Environment description (EN 206 (British Standards Institution, 2013))
XC1	4	4	Dry or permanently wet; indoor environments with low humidity or fully submerged concrete.
XC2	3.5	3	Wet, rarely dry; surfaces subject to long-term water contact, typically foundations.
XC3	2.2	1.8	Moderate humidity; indoor environments with moderate or high humidity, rain sheltered external concrete.
XC4	3	2.2	Cyclic wet and dry; surfaces with water contact outside XC2 applications (e.g., unsheltered external concretes).

Table A6

Material carbon footprint (GWP, kg CO₂-eq. per kg of material) used for impact assessment of each concrete mix. Data is from ecoinvent database ([Swiss Centre for Life Cycle Inventories, 2022](#)) unless otherwise specified. Data for superplasticizers is per kg of solution, given dilutions in [Table A3](#).

Material	GWP (kg CO ₂ -eq. kg ⁻¹)	Source
Clinker	0.86	Ecoinvent, Market for clinker, CH
Limestone	5.4×10^{-3}	Ecoinvent Market for limestone, crushed, washed, CH
Gypsum	8.2×10^{-3}	Ecoinvent, Market for gypsum, mineral, RER (based on CH gypsum quarry operation)
Calcined clay	0.314	Data from (Miller et al., 2018), European production
Sand	5.4×10^{-3}	Ecoinvent, Market for sand, CH
Aggregate	5.4×10^{-3}	Ecoinvent, Market for gravel, round, CH
Water	0.00	Ecoinvent, Market for tap water, CH
MasterGlenium SKY	0.57	EPD, Master Builders Solutions
MasterCO ₂ re	0.62	EPD, Master Builders Solutions
MasterPel	1.37	EPD, Master Builders Solutions
Master X-Seed	0.73	EPD, Master Builders Solutions

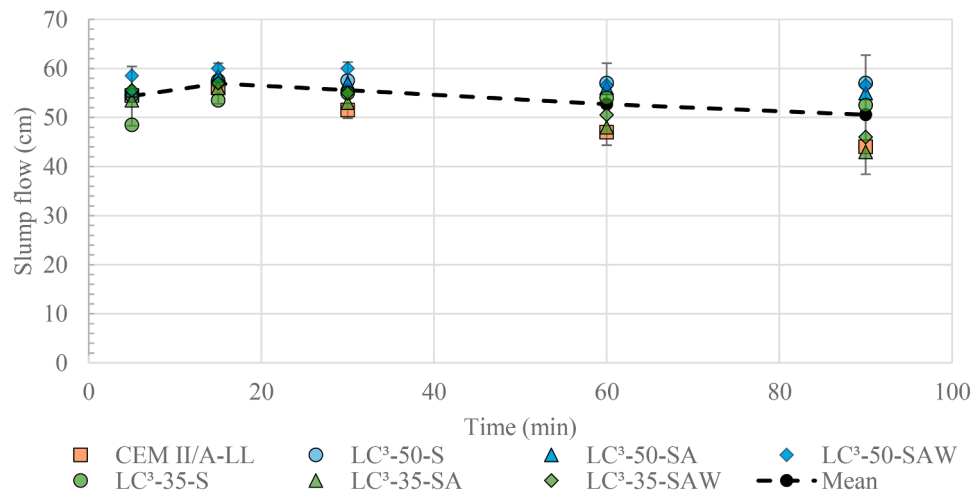


Fig. A1. Flow table test results for the final dosage of superplasticizer in each concrete mix: CEM II/A-LL (orange), LC³-50 (blue) and LC³-35 (green). Different superplasticizer addition to LC³ mixes are differentiated by dot shape: superplasticizer only ('S' mix, circle), superplasticizer and accelerator ('SA' mix, triangle), and superplasticizer, accelerator and water repellent ('SAW' mix, diamond). Error bars demonstrate 2 standard deviations from the mean value (dotted line).

Data availability

Data will be made available on request.

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